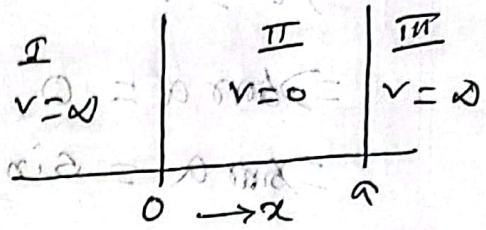
BUZON-50

(II) Given that,

$$\psi = A\sin rx + B\cos sx \quad \text{---} \quad (1)$$

where, A, B, r, s are constant



Verification: (যাচাই কর)

$\because \psi = A\sin rx + B\cos sx$

$\therefore \frac{d\psi}{dx} = Ar\cos rx - Bs\sin sx$

$\Rightarrow \frac{d^2\psi}{dx^2} = -Ar^2\sin rx - Bs^2\cos sx$

$= -(Ar^2\sin rx + Bs^2\cos sx)$

Not verified, if $r=s$ then, $\frac{d^2\psi}{dx^2} + r^2\psi = 0$

From the boundary conditions it is equivalent that

$\psi = 0$ at $x = 0$ and also at $x = a$

Using the boundary conditions, from eq. (1), we get –

(i) When $x = 0$

$0 = A\sin r.0 + B\cos s.0$

$\Rightarrow B\cos 0^\circ = 0$

$\Rightarrow B = 0$

$\therefore \psi = A\sin rx$ ---- (2)

(II) When $x = a \neq 0$ from eq (2) we get

$$0 = A \sin r x$$

$$\Rightarrow 0 = A \sin r.a$$

$$\Rightarrow \sin r a = 0$$

$$\Rightarrow \sin r a = \sin n\pi$$

$$\Rightarrow ra = n\pi$$

$$\therefore r = \frac{n\pi}{a} \qquad n = 1, 2, 3, \ldots$$

$A \neq 0$ because if $A = 0$ then $\psi = 0$ for all values of 'a' which is not true

and can not be zero ($n \neq 0$)

$\therefore$ Then eq (2) gives.

$$\psi = A \sin \frac{n\pi x}{a} \quad \ldots(3)$$

$$n = 1, 2, 3 \ldots$$

$$\therefore \psi = A \sin rx$$

(Showed)

Final → 2021 → Same as 2022.

Final 2020

(a) State the zero-point energy. (2)

Ans:

The point in which any particle exists as a most possible lower energy level then that point is called zero point energy.

From the energy expression of a particle of mass 'm' moving in a cubic box of length 'a' is given by -

$$E = \frac{h^2}{8ma^2}\left[n_x^2 + n_y^2 + n_z^2\right] \ldots\ldots (i)$$

where n_x, n_y and n_z are the quantum numbers of x, y and z directions respectively.

Now, zero point energy is possible for ground state having quantum numbers 1,1,1 for (n_x, n_y, n_z)

$$\text{So, } E = \frac{h^2}{8ma^2}(\vec{1} + \vec{1} + \vec{1})$$

$$= \frac{3h^2}{8ma^2}$$

This energy is called zero point energy.

(b) What is the physical interpretation of the wave function? Mention the acceptable conditions of ψ. → (4)

* Conditions have discussed.

Then,

Physical interpretation of the wave function:

In quantum mechanics, a wave function represents the state of a particle or system, and its physical interpretation is that the square of its absolute value, $|\psi|^2$, gives the probability density of finding the particle at a specific location in space at a given time; essentially, it describes the likelihood of observing a particle at a certain point, not its exact position due to the wave-particle duality concept.

(e) Brifely explain the results of the particle in one-dimensional box calculation. (5)

Ans: The Q. gives 5 marks, so the discussion of results is not enough. Hence, derivation / calculation should also be written for final exam which is noted down in Abser sir sheets.

The results of the particle in one dimensional box calculation are of interest with regard to the following two points —

(i) Unlike the classical prediction, the probability of finding the particle is not constant but is a function of x, furtherforce, the probability of finding the particle in a particular portion of the box depends upon the energy of the particle.

(ii) Another difference from classical prediction is the fact that only certain energies, related to the quantum numbers n are allowed. Zero energy is not allowed and the energy increases as n^2.

[Because $n \neq 0$, otherwise $\psi = 0$ and the solution is trivial — the probability of finding the particle is zero ($\psi^2 = 0$) and therefore it does not exist!).

$n-l-1$
$(n-1)-l$

(d) How many radial nodes does each of the following orbitals possess:

(i) 2s (ii) 4s (iii) 3p (iv) 5d → (3)

Ans:

nodes → probability of finding e⁻ is 0

	n	nodes (n-1)	Angular nodes (l)	Radial nodes
2s	2	1	0	1
4s	4	3	0	3
3p	3	2	1	1
5d	5	4	2	2

Formula: No. of radial nodes = $(n-l-1)$

or $(n-1)-l$

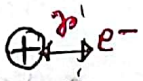
* Final → 2019 *

(a) Write the Schrodinger wave equation for the H atom in Cartesian coordinate system and explain the terms in it. → (3)

Ans:

Schrodinger equation for H atom in cartesian coordinate system:

Bohor atom model of H atom —



The potential energy of the electron of hydrogen atom depends on r — the distance between the nucleus and the electron. To bring an electron from infinity (where the influence of the nucleus charge is zero) to a distance r, the work doone will be—

$$\int_{\infty}^{r} F(r)\,dr = \int_{\infty}^{r} \frac{e^2}{r^2}\,dr = -\frac{e^2}{r}$$

And it will be the P.E of the system i.e.

P.E $V = \frac{-e^2}{r}$ for hydrogen atom

Now the general wave equation is —

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$$

Putting the value,

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{e^2}{r}\right)\psi = 0$$

which is the required equation

For H like systems (e.g. He^+, Li^{2+} etc)
The schrodinger equation will be—

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}\left(E + \frac{Ze^2}{r}\right)\psi = 0$$

Where,

∇ = laplacian operator

$= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

ψ = Wave function

h = planck constant

m = mass of particle

E = Total energy $= (T+V)$

Z = Atomic number of atom.

e = charge of e⁻

r = distance betn e⁻ and nucleus.

(a) Explain the terms 'radial distribution function'. → (2)

Ans:

The radial distribution function (RDF), often denoted as $g(r)$, describes how particles density varies as a function of distance from a reference particle in a system. It provides information about the probability of finding a particle at a certain distance r from another particle, compared to the probality expected for a completely random distribution.

Mathematically,

$$g(r) = \frac{\rho(r)}{\rho_0}$$

where,

$\rho(r)$ is the local particle density at a distance r from a reference particle

ρ_0 is the average particle density of the system.

(d) Use Slater's rule to estimate values of Z_{eff} for (i) a 4s (ii) a 3d electron in a V atom ($Z = 23$)

Ans:

(a) Effective nuclear charge (Z_{eff}):

The effective nuclear charge is the net positive charge experienced by an electron towards the nucleus in an atom. It is denoted by Z_{eff}.

$$Z_{eff} = Z - \sigma/s$$

where,

Z is the nuclear charge.

σ/s shield or screening constant.

s

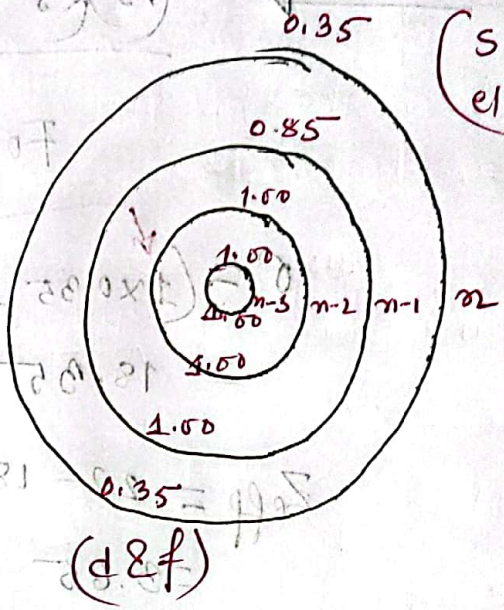
$n-1 \longrightarrow 0.85$

$n \longrightarrow 0.35$

$n-2 \longrightarrow 1.00$

Trick's for Slater's rule:

e⁻ in orbital	Per electron in orbital		
	n	n-1	n-2, n-3 etc
s or p	0.35	0.85	1.00
d or f	0.35	1.00	1.00



(s & p electron)

(d & f)

Li (3) → $1s^2$ $2s^1$ → contributes nothing

$n-1$ n

$\sigma = 2 \times 0.85 +$

$Z_{eff} = 3 - 1.7$

$= 1.3$

B(5) → $1s^2$ $2s^2$ $2p^1$ → contribute nothing

Z_{eff} for p

$\sigma = 2 \times 0.85 + 2 \times 0.35 = 2.4$

$Z_{eff} = 5 - 2.4$

$= 2.6$

Ti (22) → $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^6)\ 3d^2\ 4s^2$

For 3d

$\sigma = (1 \times 0.35 + 18 \times 1.00)$

$= 18.35$

$Z_{eff} = 22 - 18.35$

$= 3.65$

For V (23) → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$

(i) For 4s →

$\sigma = 1 \times 0.35 + 18 \times 0.85 + 3 \times 1.00$

$= 18.65$

$Z_{eff} = 23 - 18.65$

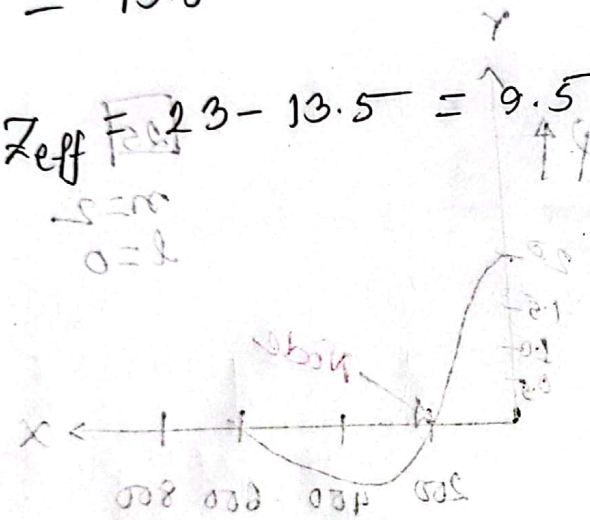
$= 4.35$

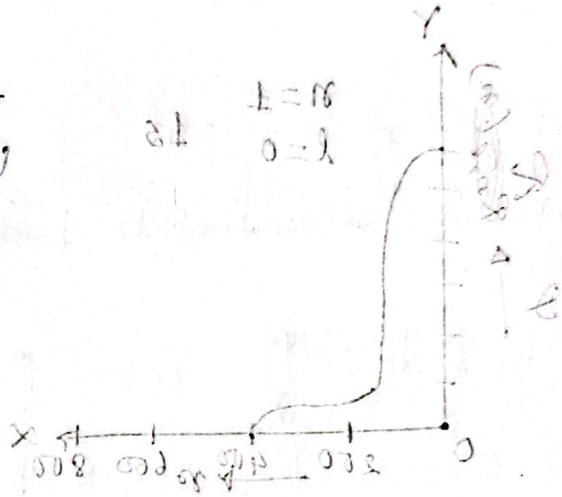
For 3d → $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$

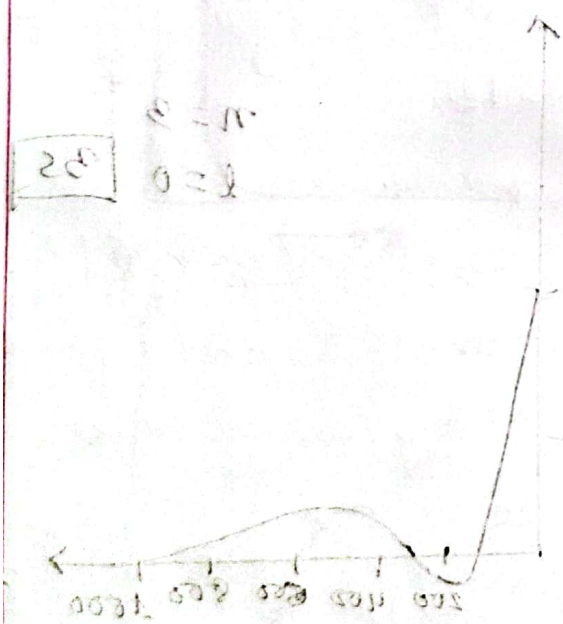
$$\sigma = (2\times1 + 8\times1.00 + 8\times0.35 + 2\times0.35)$$

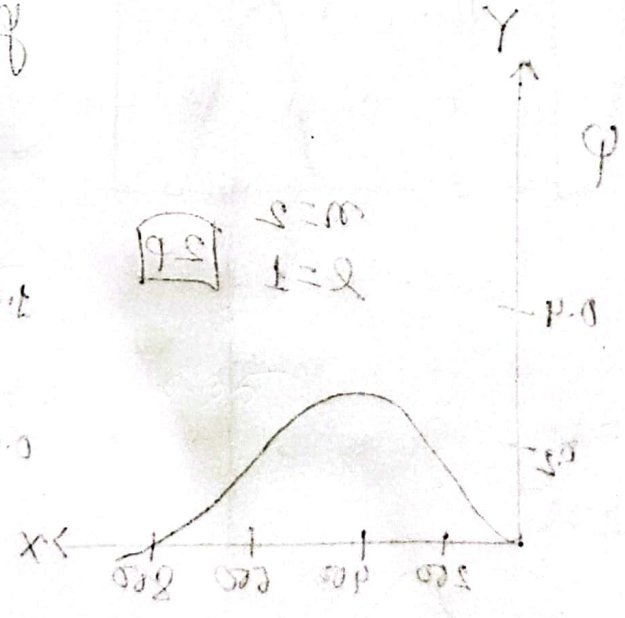
$$= 13.5$$

$$Z_{eff} = 23 - 13.5 = 9.5$$





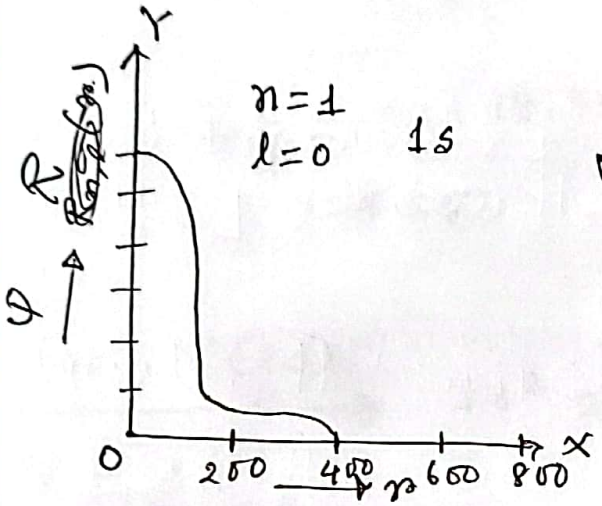


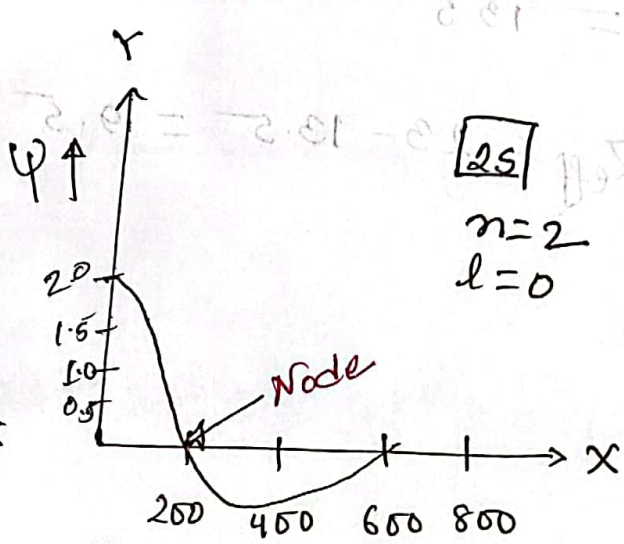


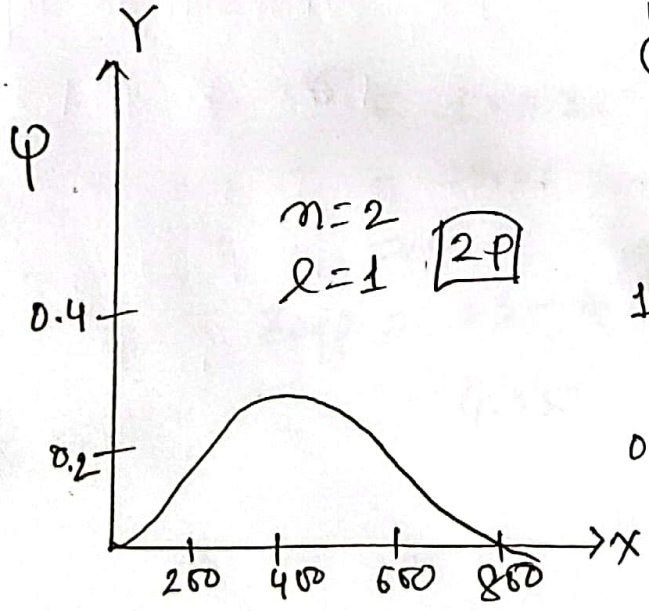
2017

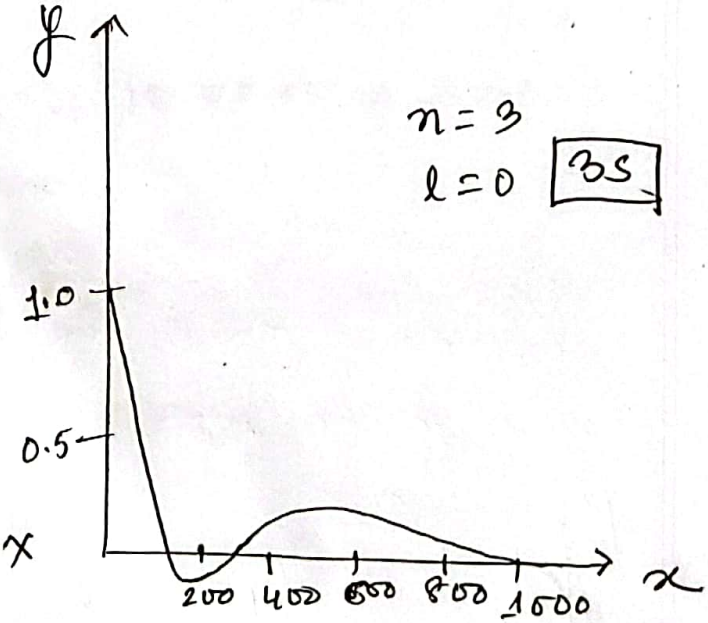
(c) Summarize and represent in figures: (i) radial part of a wave function (ii) radial probability distribution curves of the electron wave function for different electronic states of H atoms.

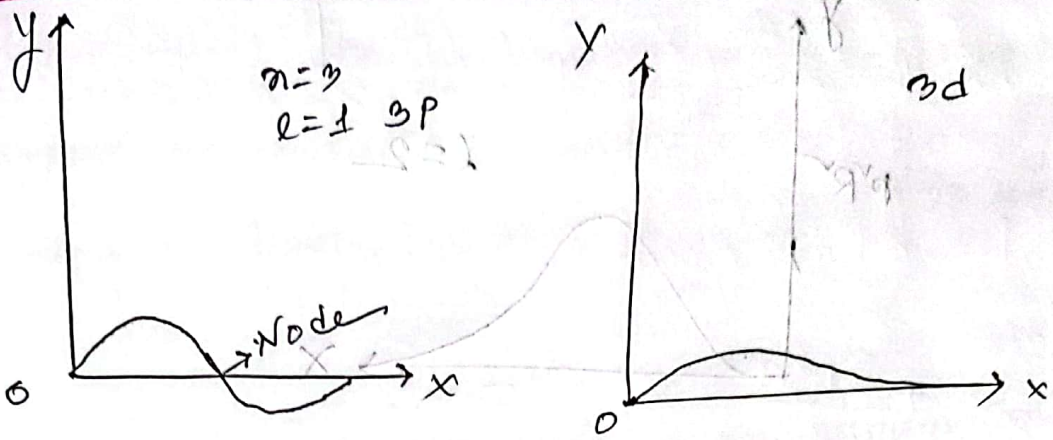
Distribution curve for radial wave function:



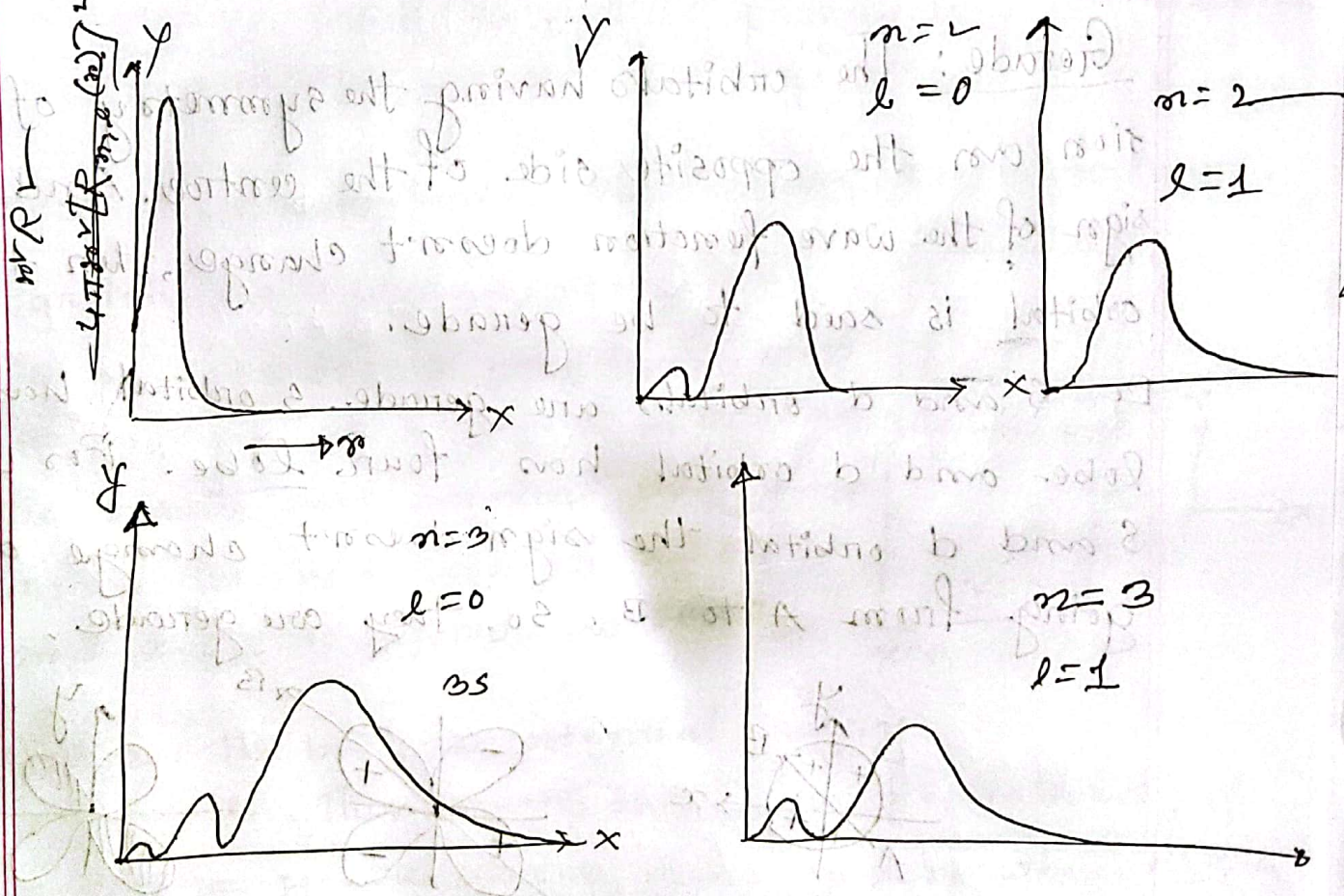


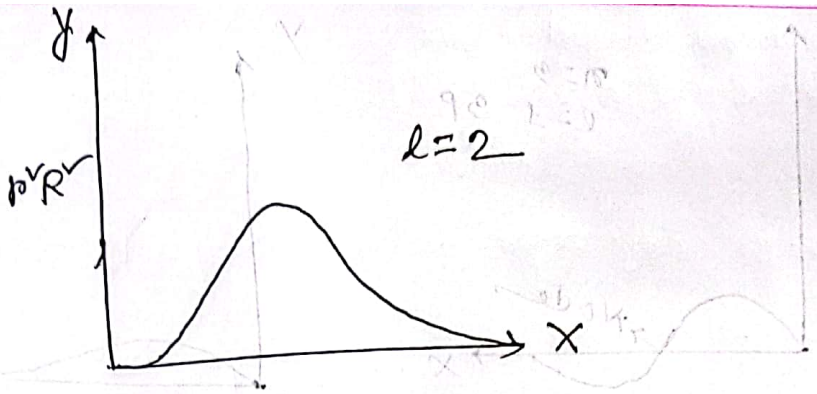






Distribution Curve for radial distribution function;



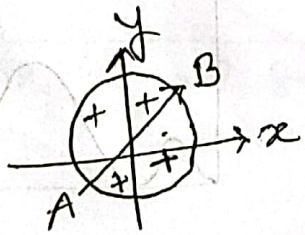


(d) What is meant by 'gerade' and 'ungerade' orbitals? Clarify with examples.

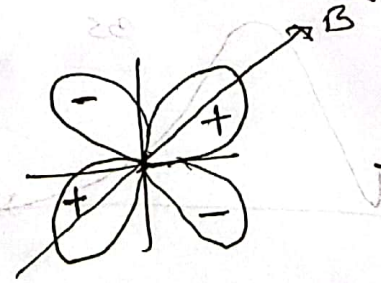
Ans:-

Gerade: The orbitals having the symmetry of inversion on the opposite side of the centre. And if the sign of the wave function doesn't change, then the orbital is said to be gerade.

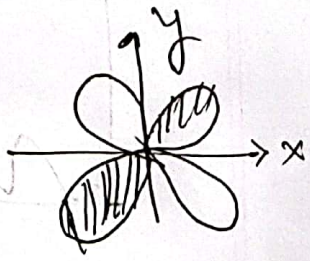
Ex: S and d orbitals are gerade. S orbital has one lobe and d orbital has four lobe. For both S and d orbital the sign doesn't change on going from A to B. So, they are gerade.



S- gerade



dxy -gerade



dxy orbital

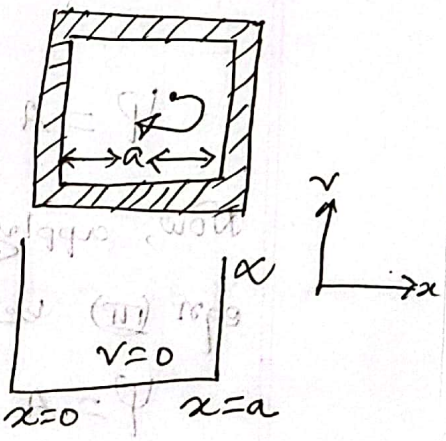
Ungerade: The unsymmetrical orbitals with respect ~~respect~~ to inversion and if the sign of wave function changes then the orbital is said to be ungerade.

(a) A particle is confined to a one-dimensional box of mass m and length a having infinitely potential walls and its total energy remains constant. Evaluate the energy values E_n of the ~~gerade~~ particle and illustrate the energy levels as well as probability distribution functions of the above mentioned box.

Ans: → আবদ্ধাবস্থার জন্যে একমাত্রিক আদর্শ ধরনের বেড়া,

Particle in one dimensional box:

Let us consider a particle of mass m moving in a straight line along the x-axis and potential energy V inside the box between $x=0$ and $x=a$ is defined as zero.



Outside the box, the potential energy is infinite. This means that, the probability of finding the particle outside must be zero when $0 > x > a$

To avoid discontinuity at $x=0$ and $x=a$, the wave

function must be zero and the boundary conditions be satisfied $\psi = 0$ at $x = 0$ and $\psi = 0$ at $x = a$.

Now, the schrodinger wave eqn for one dimension

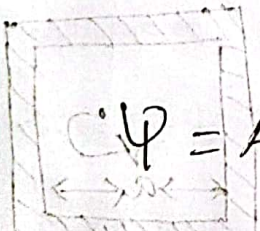
$$\frac{\delta^2\psi}{\delta x^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \quad \cdots (1)$$

Since, inside the box potential energy $V = 0$

then, (i) $\Rightarrow$

$$\frac{\delta^2\psi}{\delta x^2} + \frac{8\pi^2 m}{h^2}(E\psi) = 0 \quad \cdots (II)$$

The solution of this eqn is $\longrightarrow$



$$\psi = A \sin\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x + B \cos\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x \quad \cdots (III)$$

Now, applying the condition, $\psi = 0$ at $x = 0$

eqn (III) be comes $\longrightarrow$

$$\psi = A \sin\left(\frac{8\pi^2 mE}{h^2}\right)^{1/2} x \quad \cdots (IV)$$

~~Again, applying~~

2016

* What is the significance of schrodinger eqn?

*The wave function described by schrodinger eqn contains all possible information about the quantum system.

* Solving the eqn can determine properties like the probability of finding a particle in a certain location.

(b) What is meant by (i) Well behaved function (II) Normalization of wave function? → (3)

Ans:

Well behaved function:

A function full filling the following condition is known as well behaved function.

(i) ψ must be finite for all values of x

(II) ψ must be single valued, that is for each value of x,y must be only one value.

(III) ψ must be continuous.

(II) Normalization of wave function:

If the probability of wave finding the particle in overall space is equal to one, then the function is called normalized wave function and the process is called normalization of a wave function.

Thus,

$$\int_{x=0}^{x=a} \psi_n^2 \, dx = 1 \quad \cdots\cdots (i)$$

The schrodinger eqn for a particle in one dimensional box,

$$\psi_n = A \sin\left(\frac{n\pi}{a}\right)x \quad \cdots\cdots (II)$$

Putting this value in eqn (i) we get,

$$\int_0^a A^2 \sin^2\left(\frac{n\pi}{a}\right)x \, . \, dx = 1$$

$$\Rightarrow \frac{1}{2} A^2 \int_0^a 2 \sin^2\left(\frac{n\pi}{a}\right)x \, dx = 1$$

$$\Rightarrow \frac{1}{2} A^2 \int_0^a \left[1 - \cos\left(\frac{2n\pi}{a}\right)x\right] dx = 1$$

$$\frac{A^2}{2}\left[\int_0^a 1\,dx - \int_0^a \cos\left(\frac{2n\pi}{a}x\right)dx\right] = 1$$

$$\Rightarrow \frac{A^2}{2}\left\{[x]_0^a + \left[\frac{\sin\frac{2n\pi}{a}x}{\frac{2n\pi}{a}}\right]_0^a\right\} = 1$$

$$\Rightarrow \frac{A^2}{2}(a-0) = 1$$

$$\Rightarrow A^2 = 2/a$$

$$\therefore A = \sqrt{2/a}$$

Now, $\psi_n = \frac{\sqrt{2}}{\sqrt{a}}\sin\left(\frac{n\pi}{a}\right)x$

Where, ψ_n is called normalized wave function.

(c) Write the Schrodinger equation for a particle in one-D box and of 3-D box. → 2

Ans:

(d) Griv.

*(d) Given the energy expression for a-particle of mass 'm' moving in a cubic box of length 'a', Hence (i) calculate the zeropoint energy (ii) show that 2nd energy level is degenerate.

Ans:

From the energy expression of a particle of mass 'm' moving in a cubic box of length 'a' is given by —

$$E = \frac{h^2}{8ma^2}(n_x^2 + n_y^2 + n_z^2)$$

where, n_x, n_y and n_z are the quantum numbers of x, y, z directions respectively.

now for (I)

(I) The zero point energy is possible for ground state having quantum numbers 1,1,1 for (n_x, n_y, n_z) respectively.

So, $$E = \frac{h^2}{8ma^2}(1^2+1^2+1^2)$$

$$= \frac{3h^2}{8ma^2}$$

This is the zero point energy.

(II) Second energy level is triply degenerate.

For 2nd energy level, having a possible set of quantum numbers $\psi(2,1,1)$ for (n_x, n_y, n_z), the value of energy will be $\frac{6h^2}{8ma^2}$

similarly, for another two independent states having quantum numbers $\psi(1,2,1)$ and $\psi(1,1,2)$ for (n_x, n_y, n_z) the energy value will be $\frac{6h^2}{8ma^2}$.

SUZON-50

THE END